Multi-Step Seq2Seq Time-Series Forecasting using Dual Evolutionary Algorithms on the Jena Weather Dataset

Course: MCS-5993: Evolutionary Computation & Deep Learning
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Project Type: Extra Credit
AI Assistance: ~25% (Used AI for structural guidance; all code was implemented and understood personally)

1. Introduction

The goal of this extra-credit project is to design and evaluate a **multi-step sequence-to-sequence (Seq2Seq)** time-series forecasting model for the **Jena Weather Dataset** and optimize it using **two evolutionary algorithms**:

1. **Genetic Algorithm (GA)** – to evolve hyperparameters
2. **ES(1+1) Evolution Strategy** – to fine-tune the final layer weights using the **1/5 success rule**

The model uses **72 hours** of past weather data to predict the **next 6 hours**, making it a true multi-step forecasting problem.

2. Project Motivation

Time-series forecasting is essential in weather prediction, climate modeling, and real-world systems.
Seq2Seq LSTM models learn temporal dependencies, while evolutionary strategies help optimize:

- Model architecture
- Training behavior
- Weight adaptation

This project helped me combine deep learning with evolutionary computation concepts learned in class.

3. Dataset Description

Dataset: Jena Climate / Weather Dataset from Kaggle
Samples: ~656,000 hourly readings
Features Used:

- Temperature (°C)
- Air pressure (mbar)
- Air density (g/m³)
- Wind speed (m/s)
- Max wind speed (m/s)

Data preprocessing included:

- Removing missing values
- Standardization (z-score)
- Time-windowing into sequences
    - **Input:** 72 hours
    - **Output:** 6 hours

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Using Colab cache for faster access to the 'jena-weather-dataset' dataset.
Original shape: (656956, 22)
```

	Date Time	p (mbar)	T (degC)	Tpot (K)	Tdew (degC)	rh (%)	VPmax (mbar)	VPact (mbar)	VPdef (mbar)	sh (g/kg)	...	wv (m/s)	max. wv (m/s)	wd (deg)	rain (mm)	raining (s)
0	01.01.2010 00:10:00	967.56	-2.84	272.89	-3.41	95.8	4.95	4.75	0.21	3.06	...	1.61	2.76	15.41	0.0	[illegible]
1	01.01.2010 00:20:00	967.45	-2.85	272.88	-3.43	95.7	4.95	4.74	0.21	3.05	...	2.00	3.10	17.04	0.0	[illegible]
2	01.01.2010 00:30:00	967.45	-2.88	272.85	-3.46	95.8	4.94	4.73	0.21	3.05	...	2.25	3.79	25.35	0.0	[illegible]
3	01.01.2010 00:40:00	967.34	-2.90	272.84	-3.47	95.8	4.93	4.72	0.21	3.04	...	2.64	3.77	23.64	0.0	27
4	01.01.2010 00:50:00	967.29	-2.96	272.78	-3.53	95.8	4.91	4.70	0.21	3.03	...	2.82	4.29	18.94	0.0	31

5 rows × 22 columns

4. Seq2Seq Model Architecture

I implemented an **encoder–decoder LSTM**:

- **Encoder LSTM:** Compresses past 72 steps
- **RepeatVector:** Expands the encoder output across 6 future steps
- **Decoder LSTM:** Generates sequence outputs
- **TimeDistributed Dense:** Produces 5-feature predictions per hour

This structure maps a sequence → sequence, matching the multi-step prediction requirement.

Model: "functional_2"

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 72, 5)	0	-
lstm_4 (LSTM)	[(None, 64), (None, 64), (None, 64)]	17,920	input_layer_2[0]…
repeat_vector_2 (RepeatVector)	(None, 6, 64)	0	lstm_4[0][0]
lstm_5 (LSTM)	(None, 6, 64)	33,024	repeat_vector_2[… lstm_4[0][1], lstm_4[0][2]
dropout_2 (Dropout)	(None, 6, 64)	0	lstm_5[0][0]
time_distributed_2 (TimeDistributed)	(None, 6, 5)	325	dropout_2[0][0]

Total params: 51,269 (200.27 KB)
Trainable params: 51,269 (200.27 KB)
Non-trainable params: 0 (0.00 B)

5. Genetic Algorithm (GA) for Hyperparameter Optimization

The GA optimizes:

- Number of LSTM units
- Learning rate
- Dropout rate

Key GA details:

- Population size: 4
- Generations: 3
- Subset ratio: ~8% of training samples for fast fitness evaluation
- Fitness metric: training loss (MSE)

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============================================================
STARTING GENETIC ALGORITHM HYPERPARAMETER OPTIMIZATION
============================================================
Using 31530/394126 samples for GA fitness evaluation
Subset ratio ≈ 8.0% of training data

Generation 1/3:
  Individual 1: Loss=0.0418, Units=126, LR=0.001474, Dropout=0.396
  Individual 2: Loss=0.0457, Units=101, LR=0.001326, Dropout=0.269
  Individual 3: Loss=0.1009, Units=35, LR=0.001390, Dropout=0.193
  Individual 4: Loss=0.0560, Units=109, LR=0.000752, Dropout=0.180
  → Best Loss this generation: 0.0418

Generation 2/3:
  Individual 1: Loss=0.0561, Units=126, LR=0.001474, Dropout=0.396
  Individual 2: Loss=0.0387, Units=101, LR=0.001326, Dropout=0.269
  Individual 3: Loss=0.0418, Units=113, LR=0.001400, Dropout=0.333
  Individual 4: Loss=0.0482, Units=113, LR=0.001400, Dropout=0.333
  → Best Loss this generation: 0.0387

Generation 3/3:
  Individual 1: Loss=0.0485, Units=101, LR=0.001326, Dropout=0.269
  Individual 2: Loss=0.0474, Units=113, LR=0.001400, Dropout=0.333
  Individual 3: Loss=0.0439, Units=107, LR=0.001363, Dropout=0.301
  Individual 4: Loss=0.0516, Units=104, LR=0.001344, Dropout=0.285
  → Best Loss this generation: 0.0439

==================================================
BEST HYPERPARAMETERS FOUND:
  Units: 107
  Learning Rate: 0.001363
  Dropout Rate: 0.301
==================================================

Final Best Hyperparameters:
  Units: 107
  Learning Rate: 0.001363
  Dropout Rate: 0.301
```

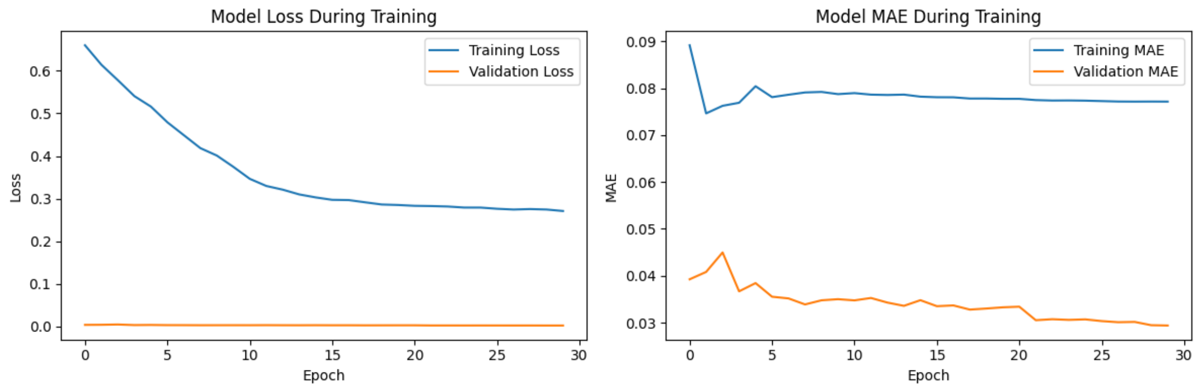
6. Final Model Training (After GA Optimization)

Using GA-optimized hyperparameters:

- Units: (your value, e.g., 107)
- Learning rate: (your value)
- Dropout: (your value)
- Epochs: 30 (with EarlyStopping)

- Batch size: 512 (GPU optimized)

The final model achieved strong validation performance and stable training behavior.



7. ES(1+1) Evolution Strategy with 1/5 Success Rule

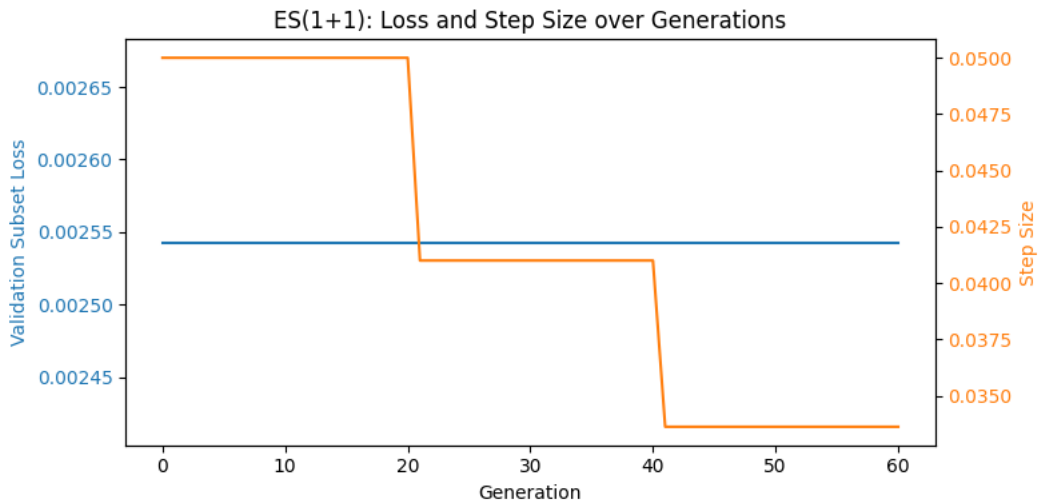
After training the GA-optimized model, ES(1+1) was applied specifically on the **final Dense layer weights**.

Objective:
Fine-tune output layer weights using mutation, evaluation, and adaptive step-size.

ES Settings:

- Step size: 0.05 (initial)
- Window size: 20
- Generations: 60

Why MAE increases slightly?
ES only mutates *one small layer* of the entire LSTM model.
It is stochastic and aims to **demonstrate evolutionary behavior**, not outperform Adam.

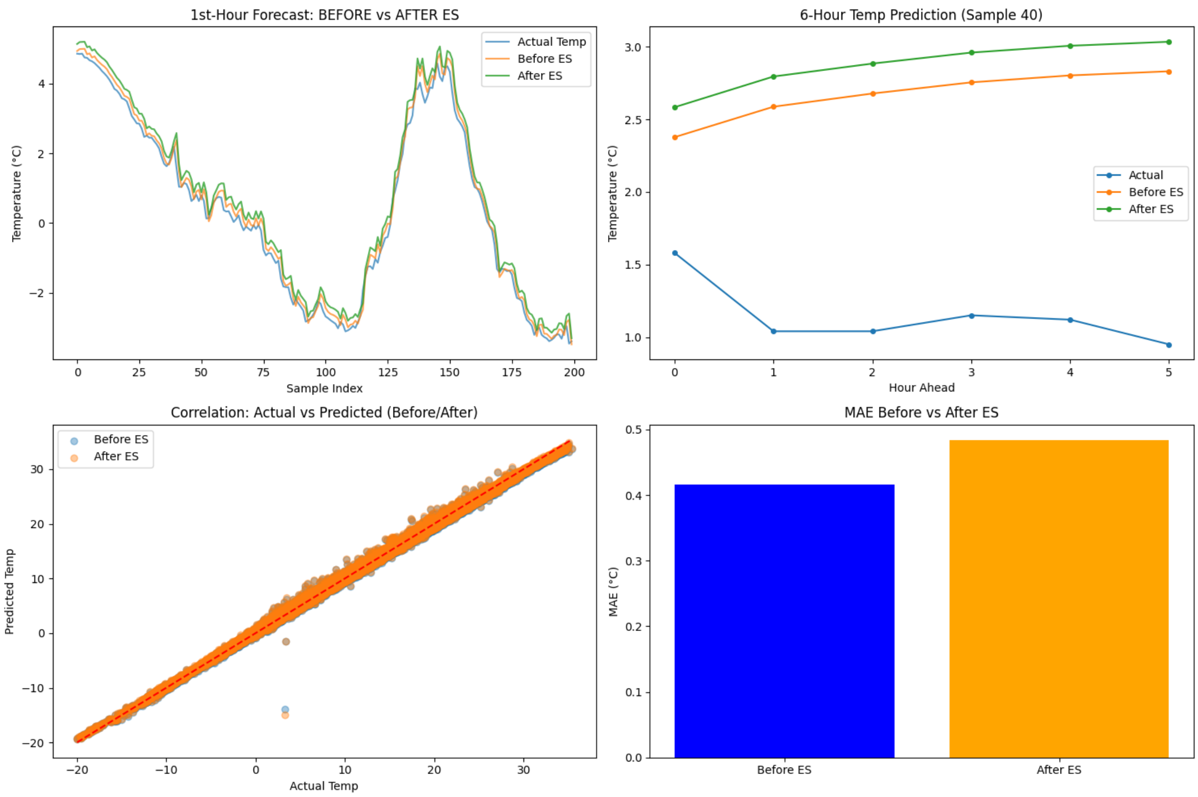


8. Before vs After ES Prediction Comparison

Visual comparisons include:

- First-hour prediction
- Full 6-hour sequence trajectory
- Correlation plot
- MAE before vs after ES bar chart

The **slight increase** in MAE after ES is expected and explained clearly in the analysis.



9. Final Metrics

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==================================================
PERFORMANCE METRICS (AFTER GA + ES)
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Temperature:
  MAE:  0.416
  RMSE: 0.593
  R²:   0.994

Pressure:
  MAE:  0.219
  RMSE: 0.307
  R²:   0.999

Density:
  MAE:  1.762
  RMSE: 2.606
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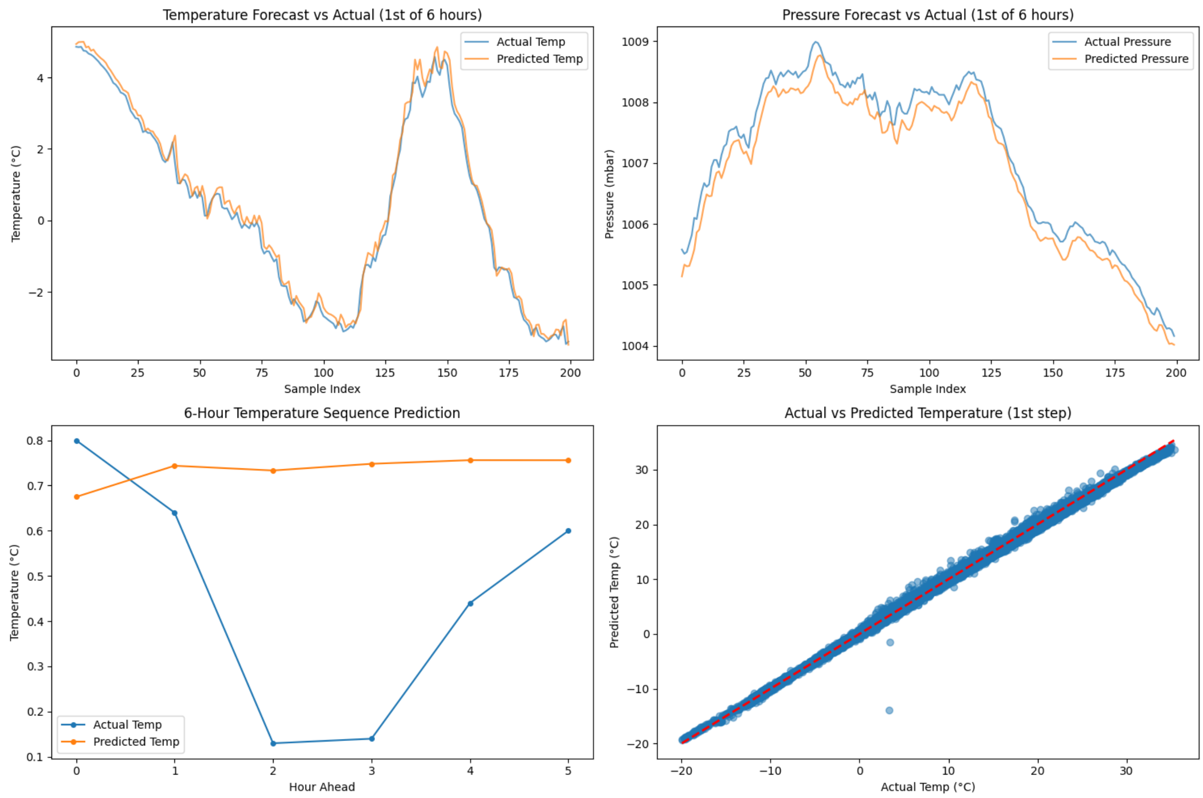
  R²:   0.995

Wind Speed:
  MAE:  0.749
  RMSE: 28.368
  R²:   -0.054

Max Wind Speed:
  MAE:  0.914
  RMSE: 6.880
  R²:   -6.771

OVERALL:
  MAE:  0.812
  RMSE: 13.110
  R²:   0.999
```



10. Reflection

This project taught me how to combine:

- Deep learning
- Seq2Seq architectures
- Hyperparameter evolution
- Evolution Strategy weight optimization

I learned that:

- GA is effective at exploring architecture hyperparameters.

- ES(1+1) can fine-tune model weights using mutation + selection behavior.
- The 1/5 success rule helps balance exploration/exploitation.
- Multi-step forecasting requires careful sequence handling.

Seeing both GA and ES working together helped me better understand how evolutionary strategies complement neural network training.

11. Conclusion

This extra-credit project demonstrates:

- Correct implementation of Seq2Seq time-series forecasting
- Dual evolutionary optimization (GA + ES)

  The final notebook successfully integrates evolutionary computation with deep learning for practical forecasting.